PS No. 11 – Project Allocation Details

Student Record Management System | BCA | Python 3

RUNGTA INTERNATIONAL SKILLS UNIVERSITY
SCHOOL OF INFORMATION TECHNOLOGY

BACHELOR OF COMPUTER APPLICATION

<table>
<tr><td>PS No.</td><td>11</td></tr>
<tr><td>Project Title</td><td>Student Record Management System</td></tr>
<tr><td>Problem Statement</td><td>Develop a system to maintain student records (roll number, name, course, marks) with the ability to add, view, search, update and delete records.
Language: Python 3 Concepts: Lists, Tuples, Dictionaries, CRUD
Note: Roll numbers must be unique and marks must be between 0 and 100. Test cases run in order on the same records.</td></tr>
<tr><td>Test Cases / Case Statements (4 cases)</td><td>
<table>
<tr><th>INPUT</th><th>EXPECTED OUTPUT</th></tr>
<tr><td colspan="2">Test Case 1: Normal Input</td></tr>
<tr><td>Menu option 1 (Add)
101, Aman, BTech, 78</td><td>Record added: 101 | Aman | BTech | 78</td></tr>
<tr><td colspan="2">Test Case 2: Boundary / Minimum Input</td></tr>
<tr><td>Menu option 1 (Add)
102, Riya, BTech, 100 (maximum marks)</td><td>Record added: 102 | Riya | BTech | 100</td></tr>
<tr><td colspan="2">Test Case 3: Invalid Input</td></tr>
<tr><td>(a) Add: 101, Neha, BTech, 70
(b) Add: 103, Ravi, BTech, 150</td><td>(a) Roll number 101 already exists.
(b) Marks must be between 0 and 100.</td></tr>
<tr><td colspan="2">Test Case 4: Different Valid Scenario</td></tr>
<tr><td>(a) Search: 101
(b) Update marks of 101 to 85
(c) Delete: 999</td><td>(a) 101 | Aman | BTech | 78
(b) Marks updated: 85
(c) Roll number 999 not found.</td></tr>
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<tr><td>Bare Minimum Requirements</td><td>1. Write the solution in Python 3 (one .py file) and run it from the console.
2. Store records using lists, tuples and dictionaries.
3. Write a separate function for each CRUD operation.
4. Ensure unique roll numbers and valid marks.
5. Display records in a neat table format.</td></tr>
<tr><td>Instruction</td><td>1. Understand the problem statement and define the problem clearly.
2. Develop the solution using proper logic and programming concepts.
3. Implement all required features and pass the given test cases.
4. Test and debug the project to ensure correct output.
5. Prepare the documentation/report and upload the code and report to GitHub.
6. Complete and submit the project within the given deadline.</td></tr>
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